

Count Vectorization Steps

step-by-step explanation of how vocabulary is created in Count Vectorization

Step-by-Step: Vocabulary Creation in Count Vectorization

1. Input Collection:
 - a. You start with a collection of documents or sentences, called a corpus.
 - b. Example:
 - i. Document 1: "I love machine learning"
 - ii. Document 2: "Machine learning is fun"
2. Text Normalization (optional but typical):
 - a. Convert all text to lowercase to ensure consistency.
 - b. Result:
 - i. "i love machine learning"
 - ii. "machine learning is fun"
3. Tokenization:
 - a. Break each document into individual words (tokens).
 - b. Document 1: ["i", "love", "machine", "learning"]
 - c. Document 2: ["machine", "learning", "is", "fun"]
4. Unique Word Identification:
 - a. Combine tokens from all documents.
 - b. Identify all unique words (this becomes your vocabulary).
 - c. From the example:
 - i. All tokens combined: ["i", "love", "machine", "learning", "machine", "learning", "is", "fun"]
 - ii. Unique words (vocabulary): ["i", "love", "machine", "learning", "is", "fun"]
5. Assigning Indexes to Words:
 - a. Each unique word in the vocabulary is assigned a fixed index.
 - b. Example:
 - i. "i" → 0
 - ii. "love" → 1
 - iii. "machine" → 2
 - iv. "learning" → 3
 - v. "is" → 4
 - vi. "fun" → 5

This dictionary of word $\rightarrow$ index is the vocabulary.

6. Result:

- a. The vocabulary is now a mapping of each unique word to an index.
- b. This vocabulary is later used to transform any input text into a count vector (word frequency vector).